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Mitigating Rapid Depressurization and Water Hammer in Subsea Water Injection Systems

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Abstract

To achieve long-term production goals, companies prioritize production efficiency and maximizing recovery from offshore fields through the strategic use of water injection. However, these systems face significant risks, including water hammer and rapid depressurization, which can damage flexible flowlines, risers, subsea equipment, and reservoirs.

The Tahiti subsea Gulf of Mexico water injection system was initially designed with three injectors and a single flowline but was later planned for an upgrade to incorporate five injectors and two extended in-field flowlines. Historically, the original system with three injectors encountered challenges like water hammer and rapid depressurization, negatively impacting both the reservoir and flexible risers. To address these challenges, a comprehensive study was conducted to identify root causes, evaluate the effects of the planned upgrade, and develop effective mitigation strategies.

Modeling flowlines in isolation fail to capture the system's complex dynamics, which are critically influenced by the interaction between injection pumps, control systems, and well tubing dynamics. As a result, the study integrated a high-fidelity dynamic simulation of the topside injection equipment with a multiphase transient simulation of the subsea flowline and well-pipe network. This holistic approach enabled a detailed analysis of historical water hammer events and facilitated the exploration of a wide range of mitigation strategies.

The integrated transient model was validated using historical data and applied to assess rapid depressurization events caused by pump stops and topsides valve closures, as well as water hammer effects during subsea flowline isolation valve (FLIV) closures. Additionally, the feasibility of increasing pump discharge pressures to address low injection rates was examined, with consideration given to steady-state operations, depressurization, and water hammer risks. Mitigation strategies, such as controlled ramp-downs of pump speeds, topside and subsea choke management, and sequential FLIV closures, were developed based on specified system maximum pressure and rate-of-change constraints. Simulation results demonstrated that these strategies effectively minimized pressure surges, ensuring system safety and reliability.

This paper highlights the application of integrated high-fidelity modeling of topside injection facilities combined with the full subsea riser, flowline, and well injection pipe network. It provides valuable insights into optimizing water injection operations and ensuring safety and efficiency in subsea systems.

Introduction

Subsea water injection systems are vital to enhancing oil recovery in deepwater developments. However, these systems face significant integrity risks during shutdown events, particularly from rapid depressurization and water hammer. These pressure transients can exceed operational design limits, potentially compromising flowline integrity, riser systems, subsea components, well completions, and reservoir performance. In severe cases, water hammer effects can propagate to the sand face, causing damage to the well completions and impacting long term injectivity.

The Tahiti field in the Gulf of Mexico utilizes a water injection system that was originally designed with three injectors and a single flowline and was later upgraded to include additional injectors and infield flowlines under the TPIC (Tahiti Producer to Injector Conversion) project. During prior operations, Chevron encountered operational challenges related to pressure transients during emergency shutdown (ESD) events, specifically rapid depressurization of topside systems and water hammer resulting from abrupt subsea valve closures. These effects have resulted in compromised well integrity and created additional risks to the safe and reliable operation of the system.

In response, Chevron commissioned a comprehensive engineering study to evaluate the water injection system's behavior under a range of shutdown conditions. The objectives are to analyze the underlying causes of pressure transients, validate a dynamic simulation model against historical field data, and develop mitigation strategies to ensure pressures and pressure derivatives remain within operational limits. In addition to evaluating the original three well configuration, the study assessed the planned expansion, including two new injector wells and a second infield flowline to ensure it would operate safely during shutdown scenarios without introducing new risks or exceeding system constraints.

While the study initially began with the three well configuration, reflecting the initial operational scope, its scope expanded as mentioned above. The early evaluations in the three well configuration were used to characterize system behavior. As the study progressed, these insights directly informed the simulation approach for the five well configuration, allowing less effective methods to be deprioritized and focus placed on scenarios with the highest operational relevance. The five well configuration reflects the current and future scope of the water injection system and presents a more conservative, higher risk operating condition due to increased flow volume and system complexity. By focusing on this configuration, the paper emphasizes the most scalable mitigation strategies while grounding its recommendations in findings applicable across both system states.

The study leverages an integrated modeling approach using high fidelity dynamic simulation to capture full transient behavior across the injection pumps, riser, flowlines, manifold, and injection wells. The model was validated against historical PI data and a transient rapid depressurization event observed in the field.

System Description

The Tahiti water injection system, located in the Gulf of Mexico, was originally designed to support three water injection wells connected via a single infield flowline. As part of the TPIC project, the system was later planned for expansion to support up to five wells, incorporating a second infield flowline and additional jumpers to improve injection coverage and reservoir support.

Figure 1 below presents a high-level process flow overview of the topsides scope using K-Spice, which includes key components involved in water injection such as pumps, control valves, and the overboard line. This figure highlights the main control elements and flow routing logic involved during steady-state injection and shutdown conditions.

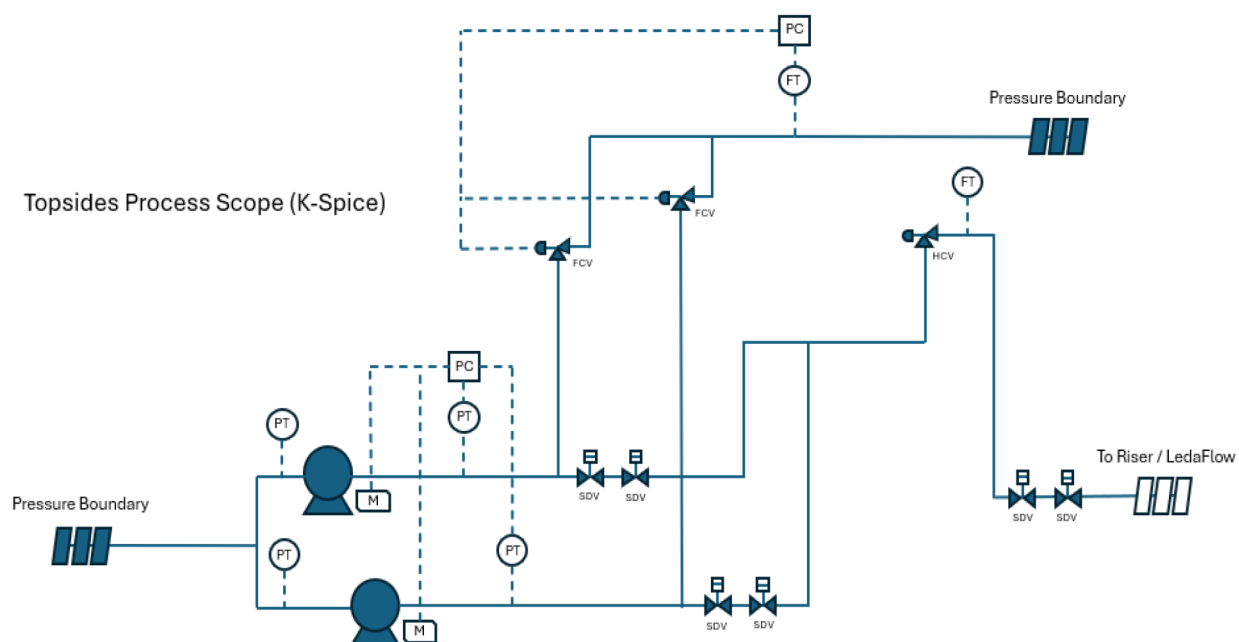


Figure 1—Model Overview of Topsides Process Scope

Figure 2 illustrates the subsea scope modeled using LedaFlow, encompassing the riser, infield flowlines, subsea manifolds, FLIVs, well jumpers, and injection control valves (ICVs). This simplified schematic is intended to provide a general layout of the dynamic modeling domain, capturing all relevant subsea equipment evaluated in the simulation.

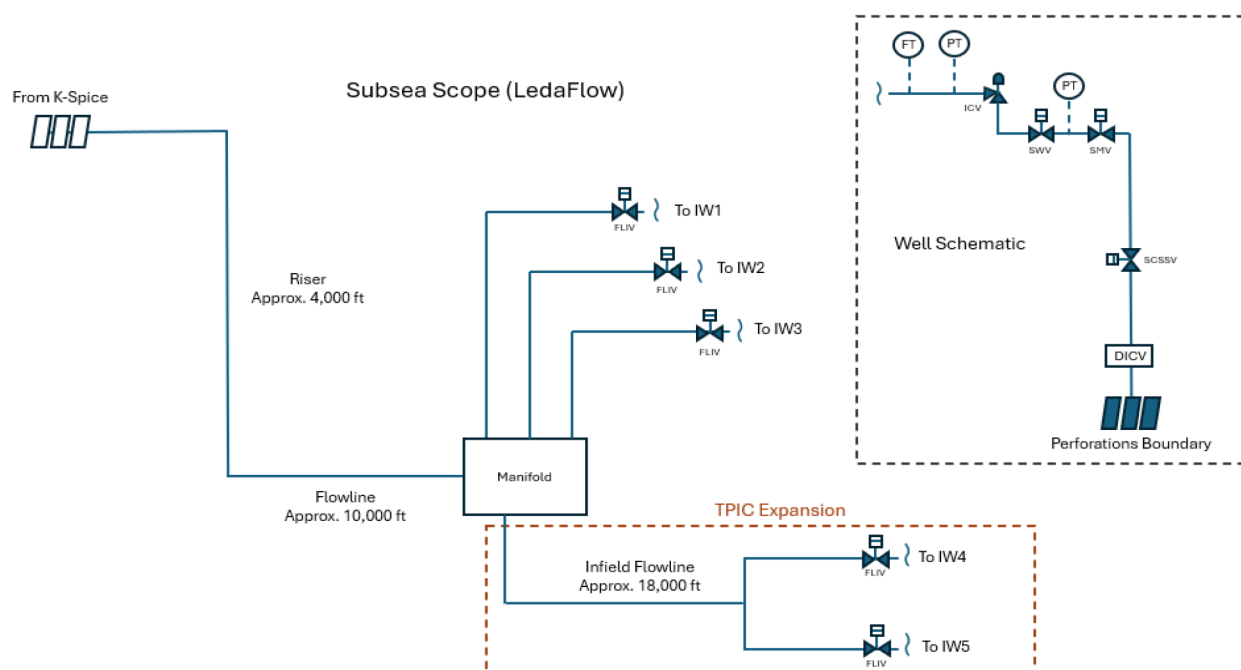


Figure 2—Model Overview of Subsea Scope

Finally, the topside and subsea models are seamlessly integrated using KDI's proprietary LedaLink, providing a single dynamic model encompassing the whole system under study.

Original System Configuration

The baseline configuration includes:

- Three water injection wells routed through a single flowline.
- Two topsides water injection pumps (variable speed).
- Overboard choke for maintaining minimum pump flow.
- A topside injection header with isolation and control valves.
- Subsea chokes (ICVs) on each well and a downhole (DICV) valve on three of the five wells.
- FLIVs are located at key junctions.

This system had a history of pressure transient issues, particularly during emergency shutdowns where rapid depressurization or water hammer events were observed. These effects were attributed to fast-acting valve closures, abrupt pump shutdowns, and insufficient coordination between topside and subsea control sequences.

The TPIC expansion (shown in [Figure 2](#)) upgraded the system with the following:

- Two additional water injection wells.
- A second infield flowline connected to the expanded subsea manifold system.

Modeling Methodology

A fully integrated dynamic simulation approach was used to evaluate the transient hydraulic behavior of the water injection system during various shutdown scenarios. The modeling framework combined a high-fidelity process simulator (representing topside injection equipment and control logic) with a transient multiphase flow simulator (representing subsea flowlines, risers, and injection wells). This coupling enabled a holistic, time-accurate evaluation of the entire system's response to fast-evolving shutdown events such as rapid depressurization and water hammer.

This integration was critical for capturing the interdependent dynamics that govern system behavior. Specifically, it allowed the model to account for:

- Pump inertia and ramp-down behavior, including how flow decays after fuel cut-off.
- Control loop dynamics, such as overboard valve regulation and choke actuation.
- Shutdown sequencing, where the order of pump stops versus valve closures affects transient severity.
- System feedback effects, where downstream events (e.g., FLIV closure) influence upstream pressures, pump performance, and control logic.

By dynamically linking these systems, the model moved beyond static boundary conditions to enable true cause-and-effect simulation of emergency shutdowns. For instance, scenarios where topside fuel was cut prior to any flow ramp-down consistently led to pressure derivatives exceeding design limits. In contrast, simulations involving a delayed pump stop and coordinated valve sequencing effectively mitigated pressure surges and maintained system integrity.

This level of insight would not have been possible with standalone topside or subsea models. The integrated approach more accurately reflects real-world operations and delivers actionable guidance for optimizing shutdown strategies, control system response, and overall risk management.

Subsea Transient Capabilities

The subsea model was constructed to accurately simulate multiphase flow behavior during highly dynamic events such as pump shutdowns and valve closures. These transient phenomena can lead to sharp pressure surges propagating through the system. To capture these events with sufficient fidelity, several advanced features were activated in the transient solver, enabling realistic modeling of pressure wave mechanics and hydraulic responses.

At a fundamental level, water hammer pressure surges are often approximated using the Joukowsky equation¹:

$$\Delta P = \rho \cdot a \cdot \Delta V \quad (1)$$

Where:

- ΔP is the surge pressure,
- ρ is the fluid density,
- a is the sonic (or wave) velocity,
- ΔV is the change in fluid velocity

The Joukowsky equation is useful for rough estimates, as it assumes single-phase flow, instantaneous valve closure, no frictional damping and rigid pipes—this equation assumes a is constant and does not dynamically update based on pipe flexibility, transient fluid properties, or valve closure characteristics. In reality, the sonic velocity a , or wave speed², in a pipe can be calculated using:

$$a = \left[\frac{1}{\rho \left(\frac{1}{K} + c \left(\frac{D}{eE} \right) \right)} \right]^{\frac{1}{2}} \quad (2)$$

Where:

- K is the bulk modulus of the fluid,
- E is the Young's modulus of the pipe wall material,
- c is the axial restraint factor,
- D is the pipe's internal diameter,
- e is the pipe wall thickness

This relationship illustrates how both fluid compressibility and pipe wall flexibility influence how fast pressure waves travel through the system. In flexible risers and long flowlines, the ability of the pipe wall to expand slightly during transients (i.e., lower E) reduces wave speed, which in turn reduces pressure spikes. Subsea flowlines and risers are typically anchored but designed to allow axial movement, functionally similar to systems with expansion joints. While the multiphase flow simulation software allows users to specify different axial restraint conditions, we assume $c=1$, representing a configuration where axial flexibility is allowed.

The dynamic flow simulation software used in this study applies first principles physics based modeling to simulate the full transient behavior of single phase water flow. It solves the conservation equations of mass, momentum, and energy dynamically, accounting for valve closure characteristics, frictional losses, the slight compressibility of water, and pipe wall deformation to accurately capture pressure wave propagation during shutdown events. Key model features for this study include:

- Flexible wall modeling, using a Young's modulus of 200 GPa, to capture riser and pipe deformation,

- Speed of sound timestep limiter, ensuring transient waves propagating near local sonic velocity are resolved,
- Axial restraint flexibility option via the anchoring coefficient c .

Model Validation

To ensure credibility in predictive applications, the integrated dynamic model was validated using field data from the PI historian under both steady-state and transient conditions. The validation process began by aligning the model with the steady-state operating conditions immediately preceding a known rapid depressurization event. Once an accurate baseline was established, the model's transient response was assessed by comparing key variables against recorded plant data, ensuring consistency across both operating regimes.

Steady-state Benchmarking

Process variables across the topsides and subsea systems, including pump discharge pressures, flow rates, and wellhead pressures, were compared to field measured data from PI. [Table 1](#) summarizes the comparison, showing good agreement across all major points, with deviations well within engineering tolerance.

Table 1—Steady-state Benchmark – Steady State Comparison Between Model and Measured Field Data from PI Historian

Description	Field Value	K-Spice Value	Units	% Error
Topsides				
Pump1 Discharge P (downstream HCV)	7781	7792	psi	0.14
Flow to Subsea	77909	77477	BPD	0.14
Flow Overboard	10396	10396	BPD	0.00
Injection Well 1	Field Value	K-Spice Value		% Error
Injection Rate	5643	5536	BPD	1.90
P Upstream Wellhead ICV	9235	9130	psi	0.14
P Upstream Downhole ICV	17613	17539	psi	0.42
P Downstream Downhole ICV	16565	16451	psi	0.69
Injection Well 2	Field Value	K-Spice Value		% Error
Injection Rate	16054	16060	BPD	0.04
P Upstream Wellhead ICV	9206	9123	psi	0.90
P Upstream Downhole ICV	19007	18980	psi	0.14
P Downstream Downhole ICV	17274	17269	psi	0.03
Injection Well 3	Field Value	K-Spice Value		% Error
Injection Rate	6698	6693	BPD	0.07
P Upstream Wellhead ICV	9233	9130	psi	1.12
P Upstream Downhole ICV	Not Available			
P Downstream Downhole ICV	Not Available			
Injection Well 4 (PIC Expansion)	Field Value	K-Spice Value		% Error
Injection Rate	25457	25142		1.24
P Upstream Wellhead ICV	8104	8107		0.04
Injection Well 5 (PIC Expansion)	Field Value	K-Spice Value		% Error
Injection Rate	24100	24043		0.24
P Upstream Wellhead ICV	6901	6914		0.19

Rapid Depressurization Benchmarking

To evaluate the model's ability to capture system transient behavior, a benchmarking scenario was developed using both initial conditions and a shutdown event sequence derived from historical field data. As shown in Table 2, the simulation was initialized using steady-state values representative of a stable injection period in which all system components—including the injection pump, overboard valve, HCV, and subsea chokes—were operating under normal conditions. This particular steady-state period before shutdown was selected because it provided the highest quality and most complete dataset available for comparison.

Table 2—Transient Benchmark – Snapshot of the Simulation Before Transient Event

Description	Field Value	K-Spice Value	Units	% Error
Topsides				
Pump1 Discharge P (downstream HCV)	8459	8425	psi	0.40
Flow to Subsea	32124	31285	BPD	2.60
Flow Overboard	10462	10462	BPD	0.00
Injection Well 1				
	Field Value	K-Spice Value		% Error
Injection Rate	5047	5080	BPD	0.65
P Upstream Wellhead ICV	10321	10259	psi	0.60
P Upstream Downhole ICV	19717	19696	psi	0.10
P Downstream Downhole ICV	NA	10152	psi	NA
Injection Well 2				
	Field Value	K-Spice Value		% Error
Injection Rate	27180	26160	BPD	3.75
P Upstream Wellhead ICV	10273	10254	psi	0.18
P Upstream Downhole ICV	18882	18804	psi	0.41
P Downstream Downhole ICV	18088	17991	psi	0.53

Once the initial conditions in the simulator were aligned with field data within an acceptable tolerance, the subsequent transient sequence was simulated to mirror the operational events observed during the actual shutdown. At approximately 90 seconds into the simulation, the injection pump was ramped down to about 80% of its operating power. Simultaneously, the overboard valve began closing over a 60-second interval. Following this, the high-capacity well (Well 2), which had an initial injection rate of approximately 27,000 BPD, was ramped down by gradually closing its subsea choke over a 45-second period, during which the topsides hydraulic choke valve (HCV) was also fully closed. Although the specific rationale behind this shutdown sequence was not documented, replicating it as observed in the field was essential for benchmarking the model's accuracy under transient conditions.

As shown in Figure 3, the simulation model closely replicates both the magnitude and rate of change observed in the riser top measured pressure data. The onset of the depressurization event is captured with high fidelity, and the overall transient response, including the timing and decay slopes, demonstrates strong alignment between field behavior and model output. This agreement supports the model's capability to accurately reproduce complex shutdown sequences and associated hydraulic transients.

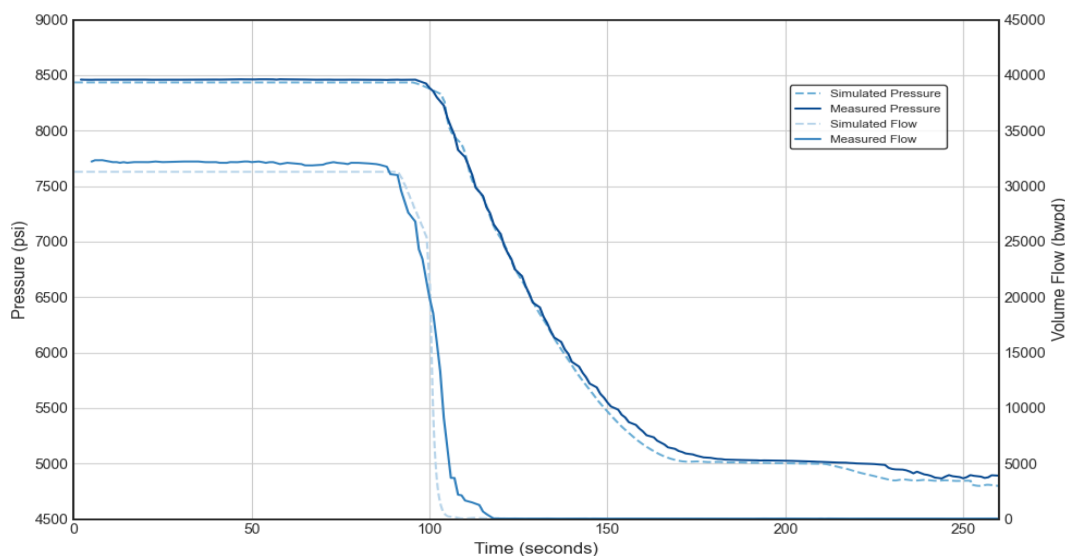


Figure 3—Transient Benchmark – Comparison of Simulated and Measured Pressure and Derivative During Rapid Depressurization

Additionally, the comparison between simulated and measured flow rates for both Well 1 and Well 2, given in Figure 4, shows strong agreement in terms of overall rate of change during the transient event. The simulated profiles successfully replicate the ramp-down behavior and relative differences between the two wells, reflecting the expected system response during the depressurization sequence. It should be noted that the frequency and consistency of the measured PI data are not uniform throughout the duration of the event. In certain sections, the measured data appears sparse or delayed, which may obscure the true dynamics of the flow rate signature. This limitation in temporal resolution should be considered when interpreting discrepancies between simulated and measured values. Nonetheless, the alignment in general shape and timing supports the model's ability to capture key flow behaviors.

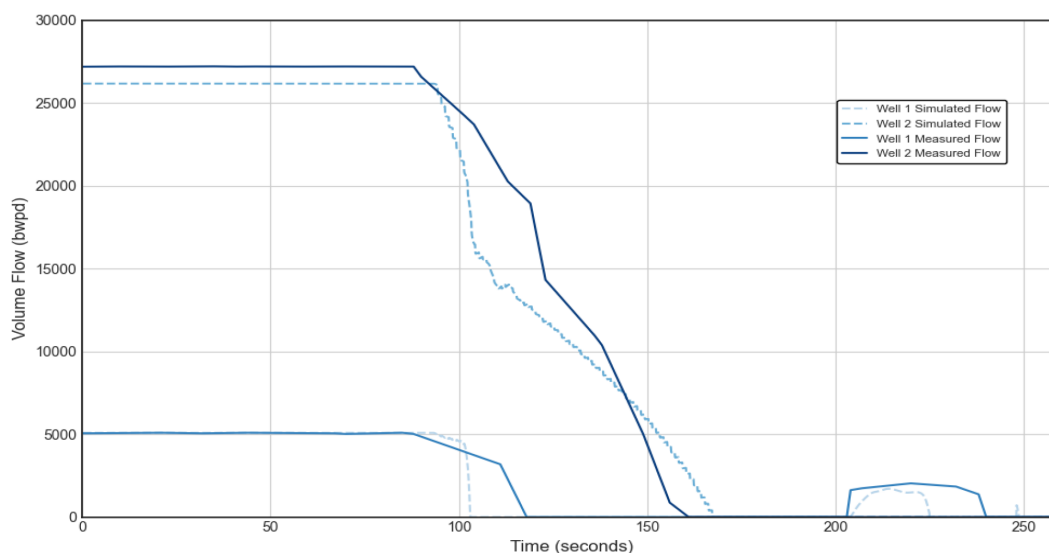


Figure 4—Transient Benchmark – Comparison of Simulated and Measured Well Flowrates During Rapid Depressurization

Rapid Depressurization

Rapid depressurization events in subsea water injection systems can be triggered by a variety of shutdown mechanisms, such as fuel gas cut to pumps, abrupt pump trips, or closure times of topside valves.

These events generate sharp pressure transients, particularly in the flexible riser and well tubing, that can exceed system design limits if not properly managed. Given the range of potential trip scenarios, shutdown sequences, and valve timing combinations, it's important to understand which factors most strongly influence the severity of depressurization.

This section aims to break down and isolate the impact of specific shutdown variables, such as pump behavior, topsides SDV stroke time, and injection pump discharge pressure, while holding other conditions constant. The purpose is to identify which elements significantly affect pressure decay and which have marginal influence. Simulations were performed across both the three-well and five-well configurations, with early sensitivity testing helping to shape the mitigation strategies later applied to the expanded system. All transient results presented below correspond to the five-well configuration.

Varying Boarding Valve Stroke Time Sensitivity

This evaluation focused on assessing whether adjusting the stroke time of the fastest closing topsides SDV downstream of the injection pumps could influence the severity of depressurization during shutdown. The intent was to determine if slowing down the valve closure could allow a more gradual reduction in flow to the subsea system, thereby reducing the rate of pressure decay and minimizing pressure derivative spikes at the riser.

For this test, the system was brought to steady-state conditions using the updated five-well configuration. Once stabilized, the pumps were left running, and the shutdown was simulated by closing the boarding valve at varying stroke times, ranging from 10 seconds (default) to 60 seconds. The purpose of this sensitivity was to isolate the effect of valve closure timing without the added influence of pump ramp-down, over boarding or fast-stop behavior.

Simulation results in [Figure 5](#), show that increasing the valve stroke time does improve the pressure decay rate, reducing the severity of the pressure derivative spike. However, even with the longest tested stroke time allowed, the maximum pressure derivative still exceeded the design limit of $\pm 14,000$ bar/hr. This is primarily due to the shutdown valve's quick-opening characteristic, where significant flow reduction occurs within the last few percentages of valve closure. As a result, while longer closure times help dampen the transient, they are insufficient alone to ensure compliance with the depressurization rate thresholds.

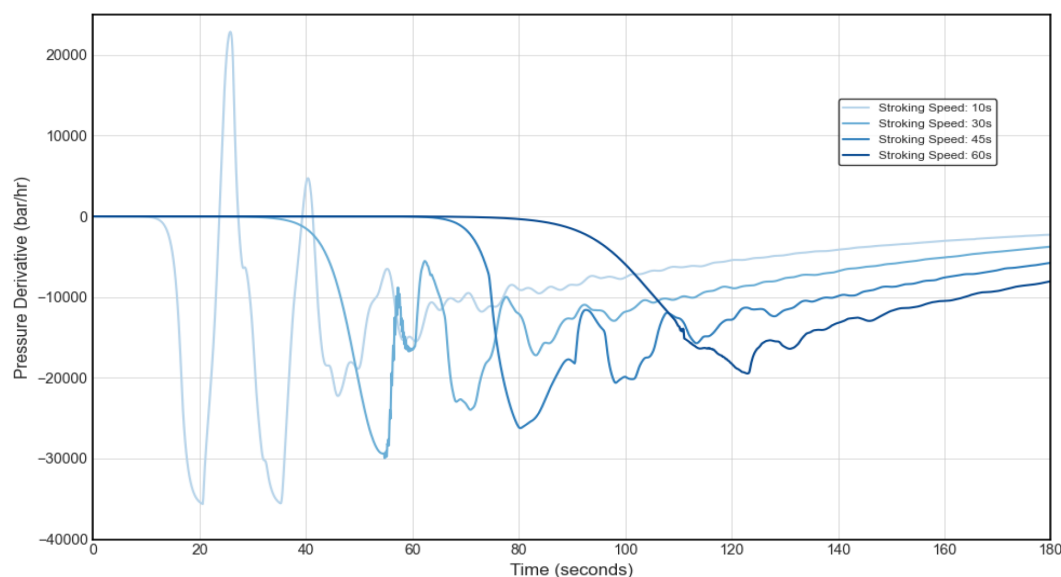


Figure 5—Sensitivity Benchmark – Comparison of Simulated Pressure Derivative Trends for Varying Boarding Valve Stroke Times.

Varying Injection Pump Discharge Pressure Sensitivity

This sensitivity study was designed to evaluate whether varying the initial injection pump discharge pressure, and as a result inlet flowrate, had any measurable effect on the severity of depressurization during an emergency shutdown of the injection pump. For each test, the model was first brought to steady-state with a target discharge pressure ranging between 7600 and 7900 psi. The corresponding injection flowrates varied slightly depending on the selected discharge pressure. Once steady-state was established, the shutdown was initiated by setting the pump discharge pressure setpoint to zero, effectively triggering a fast stop. Simulation results shown in Figure 6, highlight that varying the initial discharge pressure within this operational range had minimal impact on the severity of depressurization at the riser top. Although higher discharge pressures resulted in slightly higher starting flowrates, the differences in pressure decay rate and pressure derivative were not significant. In all cases, the pressure derivative at the riser top exceeded Chevron's threshold of $\pm 14,000$ bar/h, indicating that adjusting discharge pressure alone is not an effective mitigation strategy for this type of shutdown event.

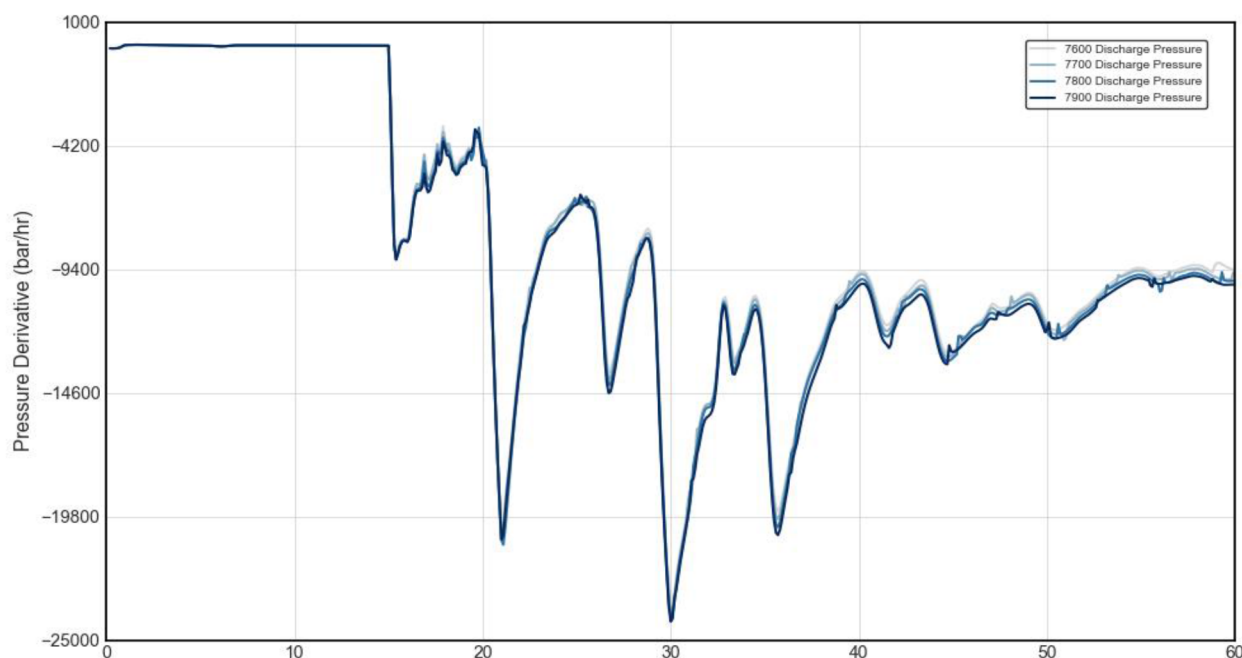


Figure 6—Sensitivity Benchmark – Injection Pump Discharge Pressure Rate of Change for Varying Discharge Pressures During Rapid Depressurization

These findings suggest that once the injection source is removed, the magnitude of the depressurization is driven more by the sudden loss of momentum and hydraulic energy in the system than by the specific steady-state discharge pressure. As such, more effective mitigation requires controlled flow reduction measures (e.g., pump ramp-down or choke manipulation), rather than minor adjustments to operating pressure setpoints.

Water Hammer

Water hammer is another critical transient risk that can arise during shutdown events in subsea water injection systems. Unlike rapid depressurization, which is typically initiated by topside pump trips or topsides valve closures, water hammer is triggered by abrupt flow stoppage at the subsea level, particularly due to the rapid closure of FLIVs during a subsea ESD. These sudden changes in momentum generate pressure surges that propagate upstream, potentially exceeding pressure safety limits and compromising riser and equipment integrity.

This section evaluates water hammer behavior through dynamic simulations focused on the five-well configuration. Several variables were examined to understand their influence on pressure spikes and derivatives during subsea ESDs. These include:

- Injection pump discharge rate, which affects the initial momentum of flow.
- Subsea choke operation, particularly the ability to ramp flow down gradually before FLIV closure.
- FLIV closure sequencing, where closing trees one-by-one rather than simultaneously.

Varying Injection Pump Discharge Rate Sensitivity

To gauge the impact of initial injection conditions on water hammer severity, a sensitivity analysis was performed to assess how varying the injection flowrate influences pressure surges during a subsea ESD. Specifically, the analysis examined whether reducing the flowrate by 5,000 to 10,000 BPD would meaningfully affect the magnitude of pressure spikes when all FLIVs close simultaneously, representing a worst-case scenario for transient propagation.

In this simulation, no explicit topside actions were taken aside from allowing the discharge pressure controller and overboard flow controller to remain in their normal operating mode, maintaining their respective setpoints. The model was initialized to a steady-state condition at each specified injection flowrate, and then a subsea ESD was triggered by simultaneously closing all five FLIVs. This setup enabled a direct comparison of water hammer severity across different flowrates without altering topside control behavior.

The results shown in Figure 7, indicate that lowering the injection flowrate helps mitigate peak riser pressures, with the highest scenario (75,000 BWPD) showing the most severe transient. However, even at the lowest tested rate (65,000 BWPD), the initial pressure derivative exceeded the $\pm 14,000$ bar/h design threshold. While lowering the flowrate provides meaningful reductions in peak pressure, it is not sufficient on its own to limit pressure derivatives within acceptable limits during sudden FLIV closure events.

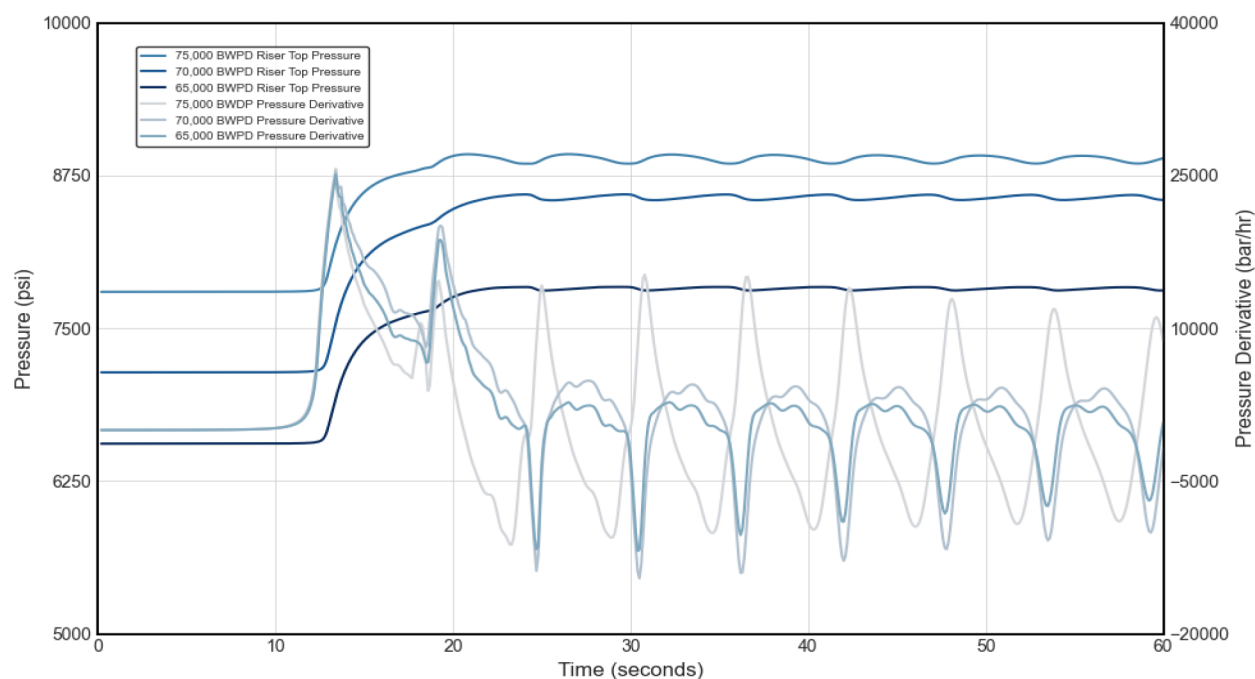


Figure 7—Sensitivity Benchmark – Comparison of Simulated Riser Pressure and Pressure Derivative During Flowrate Sensitivity Analysis for Water Hammer.

Subsea Choke Ramp-down Evaluation

To evaluate the role of flow ramp-down in water hammer mitigation, a test was performed in which all subsea chokes were ramped closed simultaneously over a 3 minute period prior to FLIV closure. The objective was to assess whether controlled flow reduction via choke manipulation could limit the severity of pressure spikes typically induced during abrupt FLIV shutdowns. All tests were conducted under current operating conditions—specifically, a steady injection pump discharge pressure of approximately 7900 psi—to match field conditions observed during prior transients. Each subsea choke was configured to close in accordance with the current control logic of 9 seconds per step. The simulation also accounted for the fact that each subsea ICV may have a different total closure time, depending on its initial opening percentage set to maintain the desired injection flowrate.

This test was compared against a baseline scenario in which all FLIVs were closed simultaneously across all five wells without any preceding choke manipulation. Again, no explicit topside actions were taken aside from allowing the discharge pressure controller and overboard flow controller to remain in their normal operation mode, holding their respective setpoints.

Figure 8 compares the ramp-down case with a baseline scenario in which all five FLIVs were closed simultaneously without prior choke manipulation. In the ramp-down scenario, water flow into the system at the point of FLIV closure was indeed lower, resulting in slightly reduced riser pressure and pressure derivative peaks. However, the improvements were not sufficient to bring either metric below thresholds, indicating that the mitigation was largely ineffective due to time constraints.

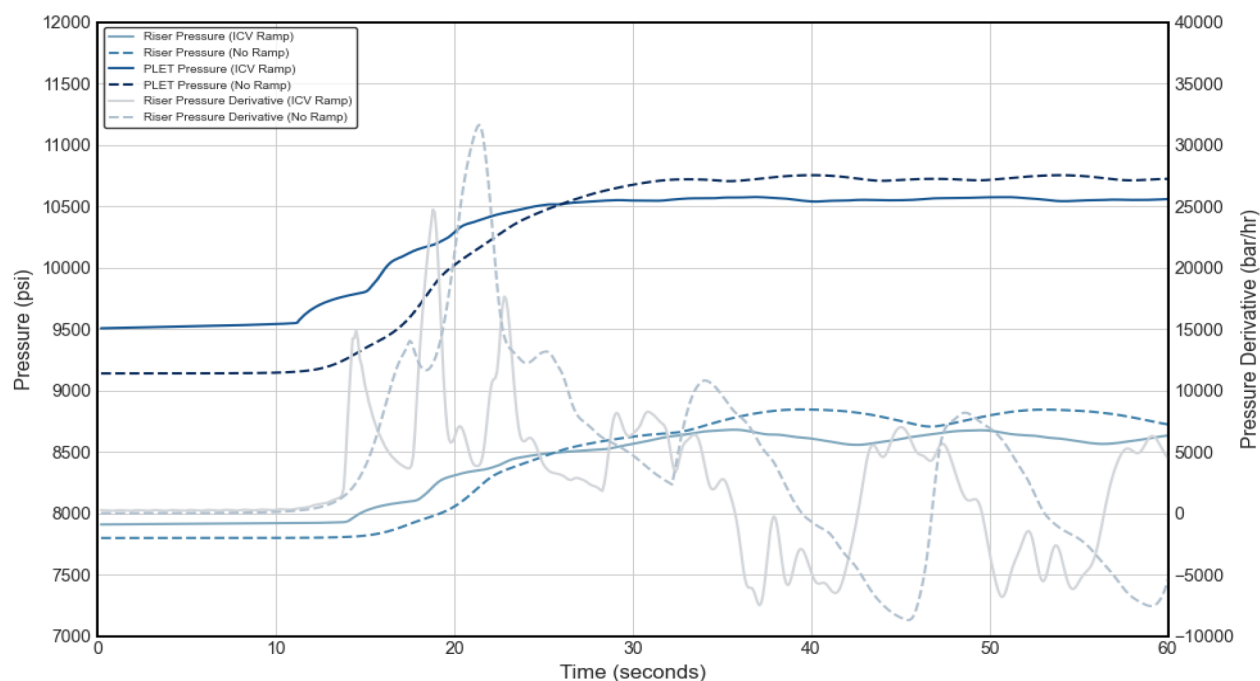


Figure 8—Subsea Choke Rampdown Response – Comparison of Pressure and Pressure Derivative for No ICV Ramp and ICV Ramp Cases

This outcome can be attributed to the stepwise nature of choke actuation where the chokes simply could not reduce flow quickly enough before FLIVs closed. Even with what is effectively a 3-minute FLIV closure delay, which represents a substantial concession in an emergency shutdown (ESD) scenario, the flowrate just prior to FLIV shutdown remained high, and the severity of the water hammer event was only marginally reduced.

These findings underscore the operational limitations of relying solely on ICV ramp-down to manage transients. While directionally beneficial, this strategy must be paired with additional shutdown measures, such as pump ramp-down and FLIV sequencing, to form an effective and practical mitigation approach.

Sequential Closing of FLIVs

A separate potential mitigation approach involves sequentially closing the FLIVs one by one, rather than simultaneously. This strategy aims to reduce the rate of injection flow decay by allowing the pump's pressure control to adjust dynamically to each individual well shutdown. The rationale is that by staggering the closures, the abrupt flow disruption that drives water hammer can be distributed over time, thereby reducing pressure surge intensity.

In this simulation, each FLIV had a 15-second stroke time and was closed in order from the lowest to the highest flowing well. A 30-second delay was applied between each closure to allow system stabilization. It is important to note that during this staggered approach, the final wells, particularly the last two FLIV closures, present a risk of exceeding the Maximum Allowable Operating Pressure (MAOP) at the riser top, since upstream pressure continues to build.

Simulation results, shown in Figure 9, highlight partial benefit. The pressure derivative remained under the $\pm 14,000$ bar/h limit during the early stages of shutdown but spiked above the threshold during the final two closures. Additionally, the riser-top pressure exceeded the Pressure Safety High (PSH) threshold of 8100 psi, indicating that neither limit was fully respected by the end of the shutdown sequence.

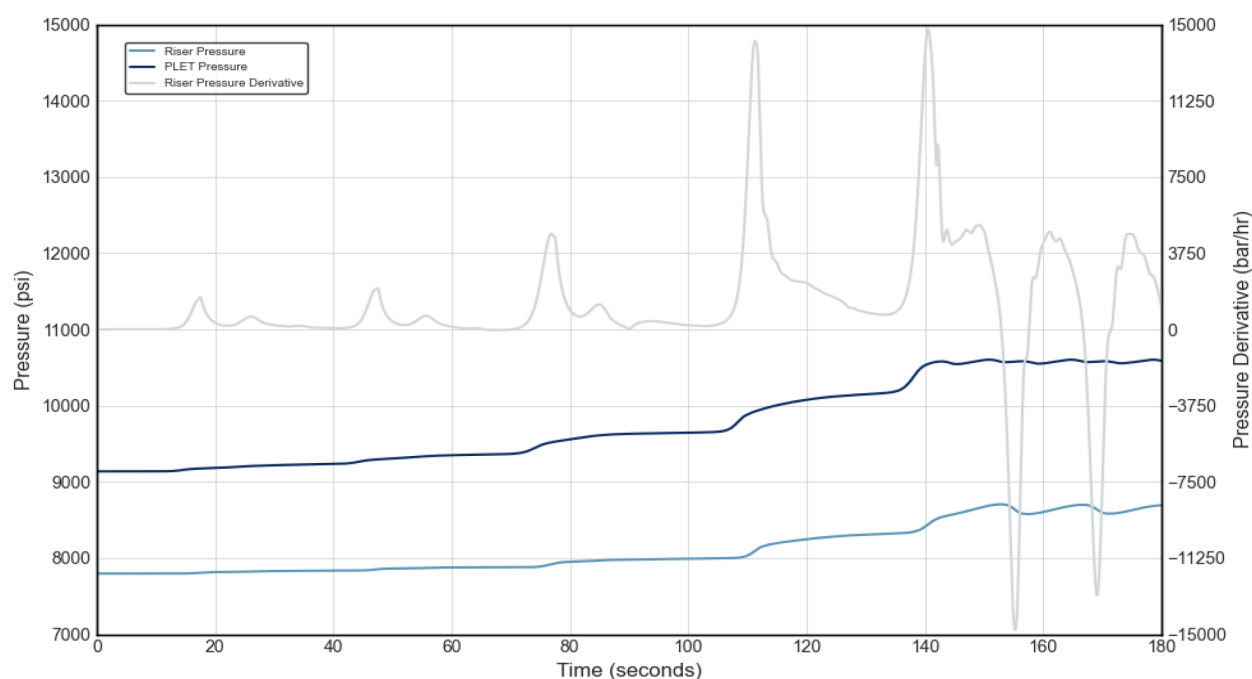


Figure 9—Sequential FLIV Closure – Riser and PLET Pressures and Riser Pressure Derivative

Another important consideration with the sequential FLIV closing strategy is the potential for increased pressure exposure at the wellhead locations (and formation sand face) of wells that remain open while others begin to shut in. As FLIVs close and system flow is redistributed, pressure can transiently rise at these points. However, as shown in Figure 10 below, the simulated wellhead pressures upstream of the subsea chokes did not exceed 10,000 psi at any location throughout the shutdown sequence. Given that this scenario was conducted without any topside mitigation actions, making it a conservative test case, it can be concluded that the sequential FLIV closure approach does not pose a risk of exceeding the MAOP at the wellheads under current design and operating conditions. While assessing sand face pressures and

their rates of change is also of interest, this analysis could not be completed within the current study but is planned as part of future work.

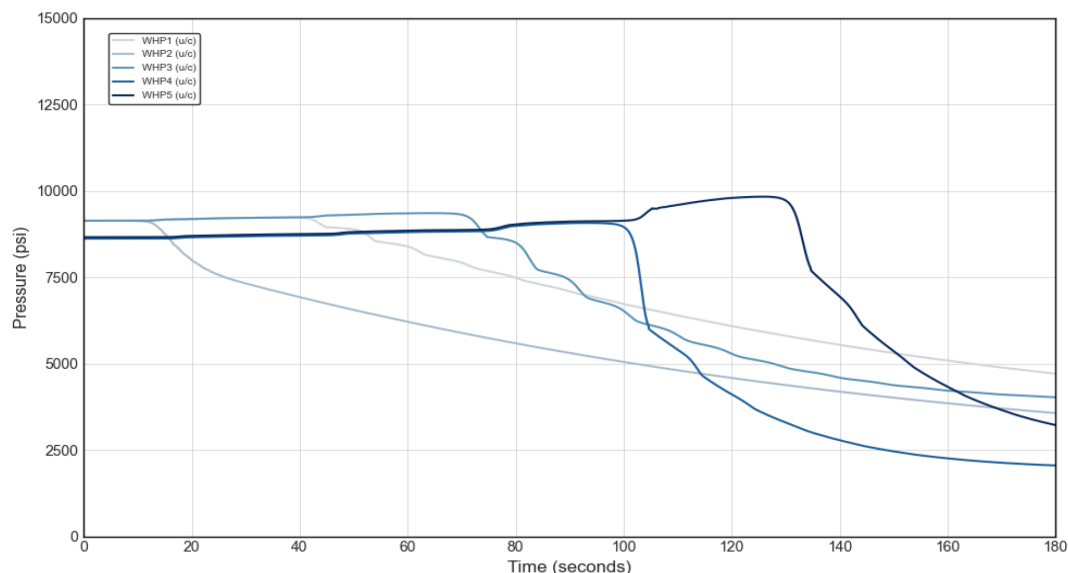


Figure 10—Sequential FLIV Closure – Wellhead Pressures During Sequential FLIV Closure

Mitigation Strategy

Balancing the mitigation of rapid depressurization and water hammer is a central challenge when designing shutdown responses for subsea water injection systems. Actions taken topsides, such as immediate pump trips or overboard valve actuation, are often required to limit depressurization rates. However, these must be carefully coordinated with subsea operations like FLIV and choke closure, which if performed too quickly can induce severe water hammer pressure spikes that propagate upstream to topsides.

This push-and-pull dynamic demands a coordinated mitigation strategy: one that avoids steep pressure decay while also controlling abrupt flow stoppages. Sensitivity studies presented above, such as variations in valve stroke times, pump discharge pressures, and subsea closure sequences, provided insight into the key contributors to pressure transients. However, these studies were exploratory in nature, aimed at understanding system response.

This section shifts toward constraint-driven recommendations, based on model outputs that evaluate shutdown scenarios against the most recent operational and design constraints. These constraints reflect allowable equipment limits, vendor design thresholds, and safety margins required to avoid damage to the flowline, injection wells, and topside equipment. Key constraints considered in developing mitigation strategies are summarized in the table below:

Table 3—Design Constraints Used for Driving Mitigation Strategies

Constraint Category	Constraint Description
MAOP - Riser	Not to exceed 8100 psi
MAOP – WHP (u/c)	Not to exceed 10000 psi
MAOP - WHP	Not to exceed 15000 psi
Pump Discharge Pressure	Pump discharge SP 7900 psi
Pressure Rate Limit	-14000 bar/hr < ΔP < 14000 bar/hr

Figure 11 presents the simulated riser top pressure and pressure derivative, along with the PLET pressure response, for the current field ESD shutdown sequence, which represents the worst case scenario observed in this study. This plot is included to highlight the severity of pressure transients generated under the existing system configuration and to establish a baseline for evaluating the effectiveness of improved, model-recommended shutdown strategies. The results clearly demonstrate that the current sequence leads to pressure derivatives that exceed acceptable thresholds, reinforcing the need for a more controlled, staggered shutdown approach to preserve system integrity and mitigate water hammer effects.

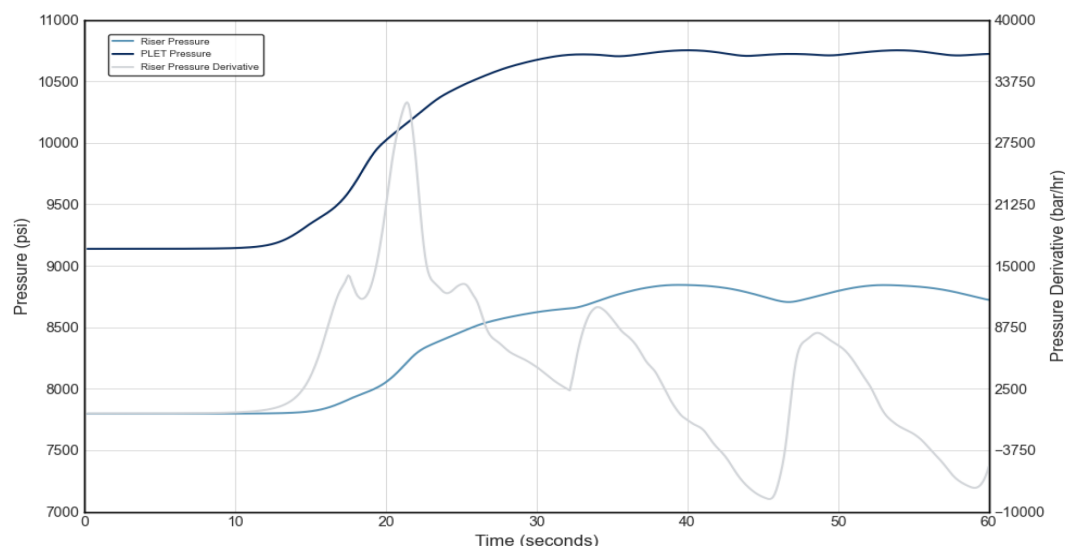


Figure 11—Current Shutdown Logic – Worst Case Scenario Showing Riser and PLET Pressures and Riser Pressure Derivative

Recommended Strategy

This strategy integrates all mitigation elements into a staged and coordinated shutdown sequence, guided by insights gained from the sensitivity analyses conducted in the Rapid Depressurization and Water Hammer sections above. It begins with pump ramp-down and partial overboard valve opening to absorb and redirect excess flow. Simultaneously, subsea choke valves begin controlled closure to restrict flow incrementally, enabling a gradual pressure reduction. Once the system flow has sufficiently decreased, the FLIV valves are closed sequentially, starting with the lowest-flow well and ending with the highest, to avoid a sudden collapse of riser flow and the associated water hammer effects.

Recommended Shutdown Sequence

- **At $t=0$:**
 - The pump begins to ramp down from operating point (in this simulation steady-state operating point is 5250 RPM) to 4900 RPM in 120 seconds.
 - The overboard valves open to 50% (from steady-state operating point of 10%), completing this action within 1 minute.
 - Subsea choke valves start closing at a rate of 9 seconds per step
- **At $t=60$ seconds:**
 - The FLIV valves begin to close sequentially, starting with the valve associated with the lowest flowrate. The closing order follows the flowrate ranking, with the highest flowrate valve being the last to close.

Simulation results confirm that this coordinated approach keeps both riser pressure and pressure derivatives within acceptable thresholds as shown in Figure 12, where depressurization rate limits remain within $\pm 14,000$ bar/hr.

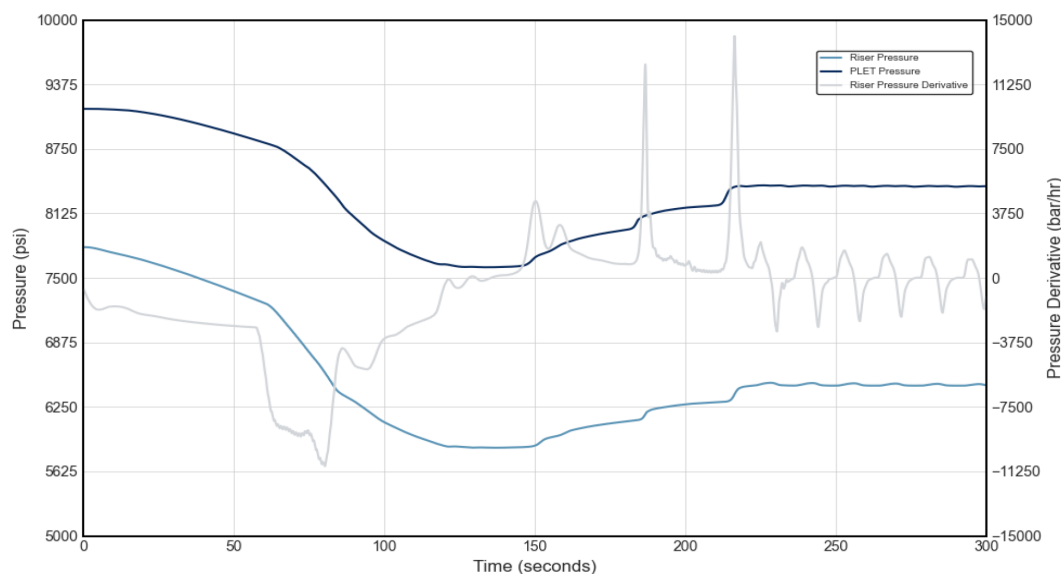


Figure 12—Recommended Mitigation Strategy – Riser and Flowline Pressures and Pressure Derivative at the Riser Top and PLET

This strategy not only mitigates topside-driven depressurization but also ensures that subsea ESD events are executed in a hydraulically safe and wellhead-protective manner (shown in Figure 13). Proper utilization of subsea chokes plays a critical role in damping transient effects by shaping the pressure profile upstream of the wells before FLIV closure.

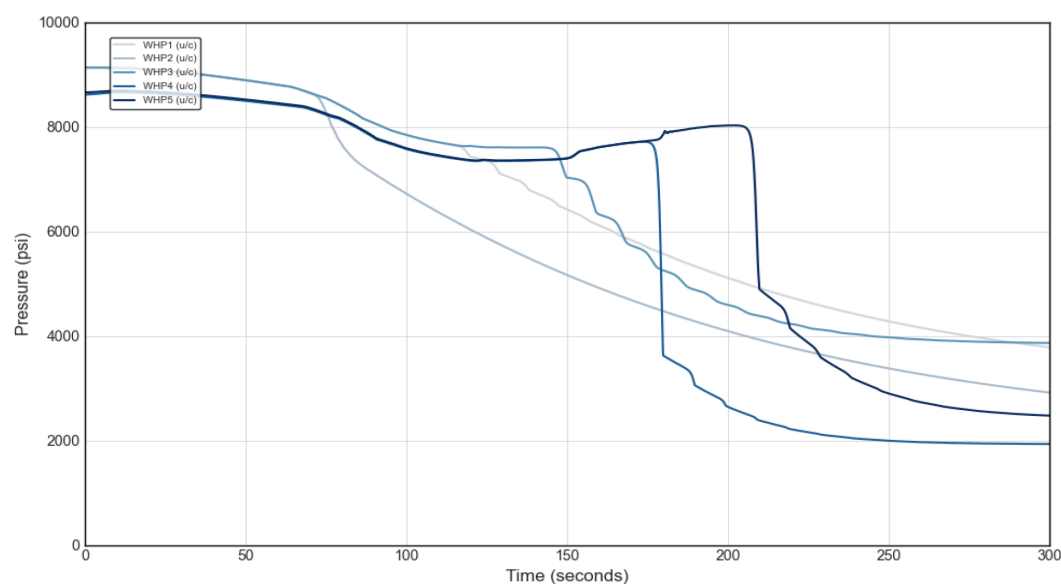


Figure 13—Recommended Mitigation Strategy – Wellhead Pressures During Mitigation Strategy.

Ultimately, the proposed shutdown sequence provides a practical, model-validated protocol to ensure system integrity under emergency conditions, enabling operations to act within defined safety margins without triggering excessive pressure surges or equipment damage.

Similar results can be achieved by using the topside choke valve to gradually restrict flow instead of relying on pump speed reduction. In scenarios where pump ramp-down is not feasible, modulating the topside choke provides an alternative means of controlling depressurization and reducing pressure transients.

Key Guidelines by Risk Category

- **For Rapid Depressurization:**

- Delay or avoid pump fast-stop shutdowns (e.g., fuel cut) whenever possible.
- If feasible, first reduce flow using pump speed adjustments or the topside manifold choke before shutting off the fuel/actuating SDVs/XVs.
- Simulations show that a 2-minute ramp-down in flow is sufficient to maintain pressure decay rates within acceptable thresholds while allowing the subsea system to start to shut down in parallel.

- **For Water Hammer:**

- Sequential FLIV closure is strongly recommended over simultaneous closure.
- Ramp subsea chokes as much as possible prior to FLIV closure to gradually reduce flow.
- Incorporate topside flow reduction measures, such as reducing pump speed or opening the topside choke valve.
- As a general rule, lower total injection flow to <10,000 BPD before closing the final FLIV to avoid exceeding pressure derivative limits.
- Ensure all discharge pumps are shut off before fully closing the last FLIV.

- **For Overall System Operation:**

- Combine subsea choke manipulation, FLIV sequencing, and topsides flow control in a synchronized, time-staggered shutdown.
- In scenarios where pump ramp-down is not feasible, similar results can be achieved by modulating the topside choke valve to restrict flow.
- This hybrid approach allows operators flexibility to meet constraints while avoiding excessive pressure transients or equipment stress.

Summary and Conclusions

This study provides a comprehensive evaluation of emergency shutdown risks and mitigation strategies in a subsea water injection system by integrating high-fidelity dynamic process simulation with transient multiphase flow modeling. By coupling topside control behavior with full-field hydraulic response, the model delivers an end-to-end, time-accurate view of system behavior under various ESD scenarios. Unlike traditional modeling approaches, this method dynamically links control actions with hydraulic responses, offering a more realistic and operationally relevant tool for shutdown strategy development. This integrated approach is a key differentiator and represents a significant advancement in how complex ESD scenarios are analyzed and mitigated.

The simulation model was validated using historical field data and enabled accurate reproduction of both steady-state and transient conditions, including a depressurization event observed in prior operations. This approach then allowed the team to confidently explore a wide range of shutdown sequences and to assess them against system design limits for MAOP and pressure rate of change.

Key Findings

- **Shutdown Risks Are Multi-Faceted:** Rapid depressurization from topside events and water hammer from subsea valve closures each pose distinct risks to system integrity. Mitigating one can exacerbate the other if not properly coordinated.
- **Single-Factor Changes Are Insufficient:** Adjustments such as extending valve stroke time or lowering pump discharge pressure provide only marginal benefit when applied in isolation and do not reliably bring pressure and pressure derivatives within design thresholds.
- **Sequential FLIV Closure Reduces Transients:** Closing FLIV valves in a staggered sequence, starting with the lowest flow well, helps distribute pressure changes more evenly and mitigates peak transients.
- **Choke Ramp-Down Alone Is Not Enough:** While subsea choke manipulation is beneficial, it is not sufficient on its own to reduce flow within the limited timeframe of an ESD. A broader, coordinated shutdown strategy is required.
- **Integrated Shutdown Sequence Is Most Effective:** The most effective mitigation strategy includes:
 - A 2-minute pump ramp-down,
 - Partial opening of the overboard valve,
 - Coordinated subsea choke closure if possible, and
 - Sequential FLIV valve closure beginning 1 minute post-ESD.

The recommended integrated shutdown sequence presented in this manuscript is to be implemented as part of broader efforts to improve the hydrodynamic response during ESD events. This strategy remains subject to internal alignment and regulatory approval. Key elements include updates to the subsea ESD logic, developed in coordination with the subsea control system vendor, to enable sequential FLIV valve closure. Rather than closing all FLIVs simultaneously after a fixed delay (e.g., two minutes post-topsides ESD), the revised logic will initiate staged valve closures beginning approximately one minute after the ESD event, with short pauses between each action to allow for pressure stabilization and better transient management. Additional enhancements to the topsides ESD sequence are also under consideration, such as implementing a controlled pump ramp-down and introducing a two-minute delay before topside valve closure, to further reduce the risk of rapid depressurization. These improvements are intended to be fully automated within the ESD control system to ensure consistent execution during both planned and unplanned events.

In summary, this study provides a practical framework (realistic, actionable approach) for developing shutdown strategies that enhance equipment protection and reservoir safety in complex subsea water injection systems. The findings support ongoing implementation efforts and regulatory engagement and offer a valuable reference for other deepwater developments facing similar risks associated with rapid depressurization and water hammer.

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